

Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Outline

- ① Motivation, question, headline
- ② The Spanish reform sequence
- ③ Model sketch
- ④ Outcomes and identification
- ⑤ Main results
- ⑥ Conclusion

Motivation, question, headline

Setting. Europe moved electricity trading from 60-min to 15-min. Spain: two reforms in 2025.

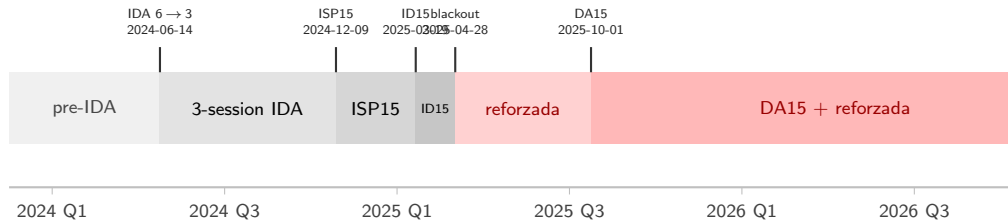
Question. Does finer granularity reshape *strategic* bidding? Who, when, which channel, market power up or down?

Headline. Granularity is a **strategic-latitude reveal**. Four empirically identified channels:

- Dispatchables widen ladders in critical hours (bid-shape DiD).
- Within-hour DA—IDA wedge SD jumps under ID15 and does not collapse at DA15 (limited per-product arbitrage).
- IDA sell-share rises sharply at ID15 for nuclear, wind, pump-storage (production-unit arbitrageurs reposition).
- Auction-stage granularity shrinks REE's ajuste cost; the post-blackout cost rise is a separate reforzada response.

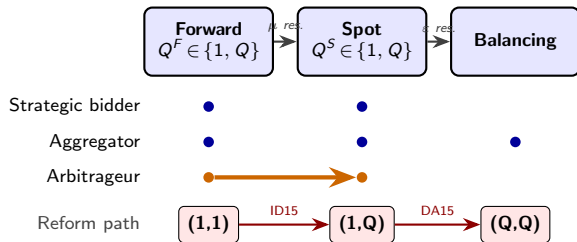
⇒ Market power **compressed** on the identified channels; not amplified.

The Spanish reform sequence



- Three granularity reforms in seven months: **ISP15** (settlement), **ID15** (intraday auctions), **DA15** (day-ahead).
- April 2025 blackout → continuous *reforzada* regime (red).
- ID15 windows: pre-blackout (clean). DA15 windows: pre starts at blackout ⇒ holds *reforzada* constant.

Model — setup



Sequential Cournot, extends Ito & Reguant (2016) with a granularity selector $Q^m \in \{1, Q\}$ at each stage. Backward induction with rational expectations.

Primitives.

- Strategic bidder: residual monopolist, marginal cost c , capacity K , no production shock.
- Aggregator: price-taker, output $y_t = \bar{y}_t + \eta_t$ with η_t forecast error (σ_η^2), bears imbalance penalty γ .
- Arbitrageur: position s , four regimes (no / full / limited / strategic, Ito & Reguant Table 1).

Shocks.

- Demand: $\mu_t \sim U[-\Delta_\mu, \Delta_\mu]$; aggregator: η_t idiosyncratic.
- Residual demand: forward $A - b_1 p_1$; spot per-period b_{2t} , average b_2 .

Model — aggregator allocates between forward and spot

Problem. Per clock-hour, split $\bar{Y}_h$ between forward MWh and per-subperiod spot MWh. Quadratic imbalance penalty $\gamma \mathcal{I}_t^2$ per delivery product on any deviation from delivered y_t :

$$\min_{\{Y^F, Y_t^S\}} p_1 Y^F + \sum_t p_{2t} Y_t^S + \gamma \sum_t E[(Y^F/Q + Y_t^S - y_t)^2] \quad \text{s.t.} \quad Y^F + \sum_t Y_t^S = \bar{Y}_h.$$

Result (Appendix ?? of thesis).

$$\frac{\partial \Delta^*}{\partial \sigma_\eta^2} > 0, \quad \Delta^* \equiv \text{MWh shifted forward} \rightarrow \text{spot at } (1, Q).$$

Intuition. A per-quarter spot lets the aggregator submit a schedule that tracks its quarter-level forecast; the hourly forward locks one schedule per clock-hour. The hedging gain is increasing in forecast-error variance σ_η^2 — the more uncertain you are, the more you want the finer scheduling tool.

Model — strategic bidder's two-tranche ladder

Problem. For each delivery product the bidder commits a ladder *before* μ_t realises. Reduce-form to two tranches (p_L, p_H) with $p_L < p_H$ and quantities (q_L, q_H) :

$$\max_{p_L, p_H, q_L, q_H} E_{\mu_t} [(p_L - c) q_L \mathbf{1}\{p_L \leq \text{MCP}_t\} + (p_H - c) q_H \mathbf{1}\{p_H \leq \text{MCP}_t\}]$$

where MCP_t depends on μ_t through the per-product residual demand $A_t - b_{2t} p$.

Result. Pre-realisation optimal spread (Appendix ?? of thesis):

$$\Delta p_{2t}^* = p_H^* - p_L^* = \frac{\Delta \mu}{2 b_{2t}}.$$

Intuition. A wider spread captures more rent in tight states (high μ_t) at the cost of foregone acceptance in slack states (low μ_t). The optimum scales *with the shock range* $\Delta \mu$ and inversely with how steep per-product residual demand is. Ramp hours $\Rightarrow$ larger $\Delta \mu \Rightarrow$ wider ladders.

Model — arbitrage and the wedge SD at (Q, Q)

Arbitrageur problem. Choose position s from forward to spot to maximise $\pi^A(s) = s(p_1 - p_2)$, possibly subject to capacity $|s| \leq K$. Four regimes (Ito & Reguant):

- (1) No arbitrage $s = 0$, scarcity wedge $p_1 > p_2$
- (2) Full arbitrage $p_1 = p_2$
- (3) Limited arbitrage $s = K$ binds, wedge $\propto 1/K$
- (4) Strategic arb. $p_1 - p_2 = (p_1 - c)/4$

At (Q, Q) , capacity is split across Q delivery products: per-product budget K/Q binds tightly. **Result:**

Wedge SD, strategic (unlimited)	$\sigma_\mu / (8 b_1)$
Wedge SD, limited per-product	$\sigma_\mu / (4 b_1)$

Intuition. Naively, once both markets are per-quarter, arbitrage should equalise prices product-by-product. It doesn't, because real arbitrageurs have finite balance sheets and now have to allocate one budget across Q separate products. Per-product capacity binds tightly $\Rightarrow$ residual wedge SD stays at *twice* the unlimited-strategic benchmark.

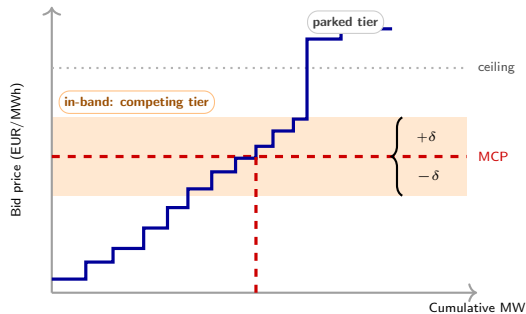
Model — comparative statics across the reform path

Outcome	(1, 1)	(1, Q)	(Q, Q)
Aggregator forward→spot shift	0	$\Delta^*(\sigma_\eta^2) > 0$	equalised
Spot ladder spread Δp_{2t}^*	—	$\Delta_\mu/(2b_{2t})$	$\Delta_\mu/(2b_{2t})$
Forward ladder spread Δp_{1t}^*	—	via Stage-1 option	$\Delta_\mu/(2b_1)$
Wedge SD (limited arb.)	0	per-period thinness asym.	$\sigma_\mu/(4b_1) + \text{thin. asym.}$
Cleared MWh / clock-hour	baseline	forward unchanged	$+c(Qb_1 - \sum_t b_{1t})/2$
Balancing residual Δ_t	$\mu_t + \eta_t + \varepsilon_t$	$\mu_t + \varepsilon_t$	ε_t
Imbalance cost $E[\Delta_t^2]$	$(\Delta_\mu^2 + \Delta_\eta^2)/3$	$\Delta_\mu^2/3$	0

Asymmetric activation of channels across the path:

- ID15 $(1, 1) \rightarrow (1, Q)$ activates the *aggregator-side* channels (quantity shift, granular-side price drop, spot-side ladder widening).
- DA15 $(1, Q) \rightarrow (Q, Q)$ activates the *strategic-bidder* channels (forward-side ladder widening, Cournot scale-up, limited per-product arbitrage).
- Welfare: dominant gain is the imbalance saving along the path; the bid-ladder widening is mostly distributional.

In-band region and bid-shape outcomes



In-band set (per curve u, d, p):

$$\mathcal{B} = \{k : |p_k - \text{MCP}_{d,p}| \leq \delta\}, \quad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}} \Delta q_j}.$$

In-band weighted mean:

$$\bar{p} = \sum_{k \in \mathcal{B}} w_k p_k.$$

Bid-shape outcomes (per curve):

$$\sigma_p = \sqrt{\sum_{k \in \mathcal{B}} w_k (p_k - \bar{p})^2}, \quad N_{\text{eff}} = \left(\sum_{k \in \mathcal{B}} w_k^2 \right)^{-1}.$$

IDA in-band sell-share (per tech t , per day d):

$$\text{SS}_{t,d} = \frac{\sum_q q_{t,d,q}^{\text{sell,in-band}}}{\sum_q (q_{t,d,q}^{\text{sell,in-band}} + q_{t,d,q}^{\text{buy,in-band}})}.$$

Bandwidth: $\delta = p_{90} - p_{50}$ of MCP per (window, market).

Identification — three specifications

Spec A: BSTS counterfactual on level outcomes (clearing prices, per-tech cleared MWh, wedge SD, IDA sell-share).

- Long pre-window (2022-01 onwards), covariates: wind_t , solar_t , gas_t , plus $\text{wind}_t \times \mathbf{1}\{\text{year} = y\}$ and $\text{solar}_t \times \mathbf{1}\{\text{year} = y\}$ for each y in the window.
- Year-interaction is essential: Spanish wind+solar capacity grew $\sim 6\times$ over 2018-2025; a flat renewable coefficient mis-fits the counterfactual and reports artificially confident effects.
- Same-calendar 2024 placebo nets recurring calendar shifts.

Spec B: per-hour-class BSTS on in-band offered energy per (tech, market, $\mathcal{X}$) cell, $\mathcal{X} \in \{\text{morning ramp, midday, evening ramp, flat}\}$. Same architecture as Spec A.

Spec C: per-curve DiD on bid shape.

$$y_{u,d,p} = \alpha_u + \beta \cdot \mathbf{1}\{d \geq T_r\} + \xi \cdot \mathbf{1}\{h(p) \in \mathcal{C}\} + \theta \cdot \mathbf{1}\{d \geq T_r, h(p) \in \mathcal{C}\} + \varepsilon$$

on $y \in \{\sigma_p, N_{\text{eff}}\}$, SE clustered by date. Identification: the within-day critical/flat contrast acts as a control, absorbing calendar variation that Spec A handles through the year-interactions.

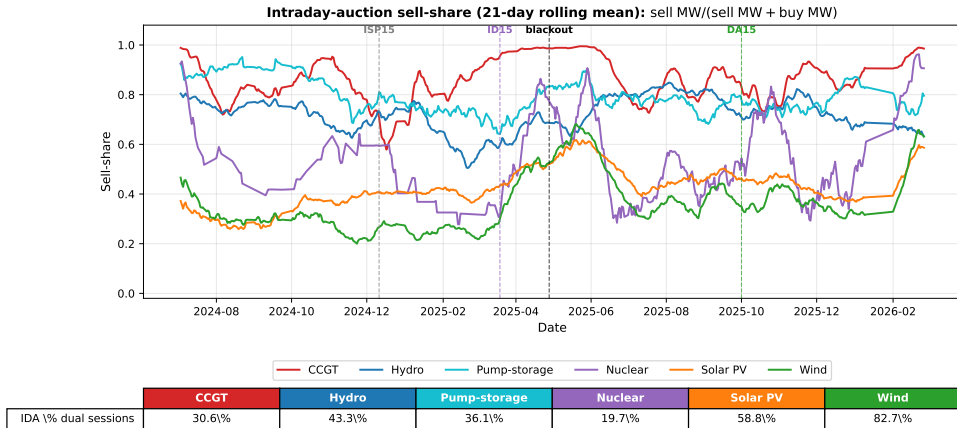
Reform windows. ID15 post-window: 2025-03-19 $\rightarrow$ 2025-04-27 (pre-blackout, 40d). DA15 post-window: 2025-10-01 $\rightarrow$ 2025-11-09 (40d). DA15 pre-window starts at the blackout $\Rightarrow$ *reforzada* held constant by design.

Result 1 — Bid ladders widen in critical hours (Spec C)

DiD θ	Tech	ID15		DA15	
		DA	IDA	DA	IDA
$\hat{\alpha}$ (EUR/MWh)	CCGT	−40.2***	11.8*	30.3***	1.4
	Hydro	0.2	87.3	43.3**	5.3***
	Pump-storage	1.1	−3.0	−6.6***	80.2
$\hat{\beta}$ (EUR/MWh per MW)	CCGT	0.27***	−0.88*	−0.17***	0.52
	Hydro	0.15	−1.50	−1.27***	−0.01
	Pump-storage	−0.01	0.07	0.02**	−1.69
$\hat{\gamma}$ ($\times 10^3$)	CCGT	−3.99***	10.0	1.29***	7.5
	Hydro	−305	−109	−5,228***	15
	Pump-storage	0.07	1.45	−0.02	0.20
N_{eff}	CCGT	−0.08	0.79	1.89***	−0.17
	Hydro	−0.24***	0.18***	0.51***	0.00
	Pump-storage	0.07	0.12*	−0.06	−0.16

- **DA15 DA CCGT/Hydro:** $\hat{\alpha} \uparrow + \hat{\beta} \downarrow + \hat{\gamma} \uparrow$ (convex). Curve translates up, flatter on average, convex bend $\Rightarrow$ low-MW MW packed cheap (Cournot scale-up) + steep high-MW tail (ladder widening). Mechanisms (iii)+(iv) at (Q, Q) .
- **ID15 DA CCGT (anticipation):** $\hat{\alpha} \downarrow + \hat{\beta} \uparrow + \hat{\gamma} \downarrow$ (concave). Aggressive climb at low MW, flatter tail. Same firm, same market: opposite curvature sign vs DA15 $\Rightarrow$ two distinct mechanisms (Stage-1 option-value, mech. v).

Result 2 — Production-unit arbitrageurs are active in IDA



- Units submit both sell and buy IDA offers. %-dual sessions: 20% (nuclear) to 83% (wind).
- Empirical confirmation of P7: arbitrageur agent active across every tech.
- DA omitted: sell-share is 1.00 for every tech (no production-unit buy side).

Result 3 — IDA sell-share rises at ID15 (Spec A)

Tech	ID15 real (pp), CI	2024 placebo
green!18 Nuclear	+39*** [+25, +53]	<i>n/a (structural)</i>
green!18 Wind	+26*** [+14, +38]	<i>n/a (structural)</i>
green!18 Pump-storage	+13*** [+7, +19]	<i>n/a (structural)</i>
Hydro	+6 [−0, +13]	<i>n/a</i>
Solar PV	+10 [−3, +24]	<i>n/a</i>
CCGT	−1 [−15, +12]	<i>n/a</i>

- Same-calendar 2024 placebo *structurally unavailable*: pre-IDA-reform (before 2024-06-14) the IDA had 6 sessions with no production-unit buy offers $\Rightarrow$ sell-share constant at 1.00 $\Rightarrow$ BSTS cannot fit.
- Falsification by descriptive break: Figure 5 (in-band sell-share rolling mean) shows the level shift right at ID15 and not at any earlier date.
- **Nuclear / wind / pump-storage rise sharply; model match** with mechanism (M3) priceable within-hour rent at (1, Q).

Result 4 — Within-hour wedge SD jumps at ID15 and does not collapse at DA15

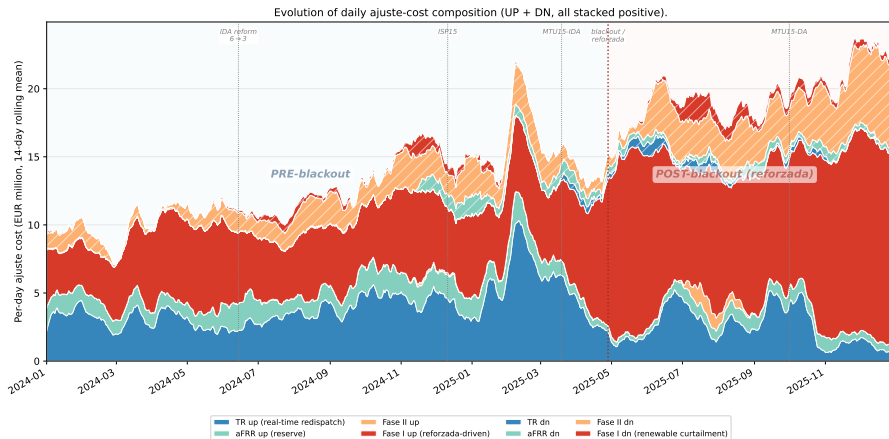
Wedge mean. $\Delta_d = \overline{p_{d,h}^{\text{DA}}} - \overline{p_{d,h}^{\text{IDA}}}$ is consistently *positive* across regimes: +1.83 pre-IDA, +0.77 ISP15-win, +2.57 MTU15-IDA pre-blackout, +0.91 post-blackout, +1.32 at DA15+ID15 (EUR/MWh). Within sampling noise of a stationary positive mean $\Rightarrow$ Ito-Reguant scarcity wedge $p_1 > p_2 > c$ under limited arbitrage.

Wedge SD (Spec A, long pre + year-int.).

Wedge SD (EUR/MWh)	Real CI	2024 Placebo CI
ID15 overall	+1.15 [+0.01, +2.23]	-0.96 [-2.07, +0.12]
ID15 critical hours	+2.12 [+0.97, +3.30]	-0.68 [-1.90, +0.48]
DA15 (no further change)	+1.12 [-0.45, +2.38]	+0.09 [-0.98, +1.15]

- ID15: significant positive jump, placebo opposite sign and not significant.
- DA15: same elevated level as ID15 (point estimate +1.12 in long-pre BSTS) — wedge SD does not revert.
- Independent 2026 placebo and ISP15 settlement-only check both confirm.
- **Model match.** Mechanism (ii) of §???: per-product arbitrage capacity binds; residual wedge SD $\sigma_\mu/(4b_1)$, twice the unlimited-strategic $\sigma_\mu/(8b_1)$ benchmark, not zero.

Result 5 — System cost: granularity shrinks ajuste; reforzada inflates Fase I



- Within-policy-window (ISP15-win → MTU15-IDA pre-blackout): total ajuste $\sim 16\text{M} \rightarrow 14\text{M}$ EUR/day. Real-time redispatch $-1/3$, aFRR up $-1/2$, down-channels also shrink. **Granularity absorbs within-hour variation upstream.**
- Post-blackout reforzada (red shading): Fase I up balloons by $\sim 5\text{M/day}$ (CCGT for voltage support); Fase II dn

Conclusion — granularity is a strategic-latitude reveal

What we identify (four channels):

- **Bid-shape widening** in critical hours — dispatchables write richer ladders where within-hour MCP variation is priceable. (Spec C DiD.)
- **Within-hour wedge SD** jumps under ID15 and does not collapse at DA15 — consistent with limited per-product arbitrage. (Spec A.)
- **IDA sell-share** rises sharply at ID15 for nuclear, wind, pump-storage — production-unit arbitrageurs reposition. (Spec A.)
- **System cost** shrinks within the pre-blackout zone — auction-stage granularity absorbs variation REE would otherwise clean up ex post. (Descriptive.)

What the model predicts and the data are consistent with but cannot identify: a level price drop at the spot reform (M2), a Cournot scale-up of cleared MWh at the forward reform (mech. iv), in-band offered-energy migration. Point estimates have the right sign; CIs are wide once renewables vary year-by-year.

Reform compresses, does not amplify, market power. Wedge mean stays at the small positive Ito-Reguant forward premium; the within-quarter strategic engagement is more competition, not more rent. Post-blackout cost rise is a separate *reforzada* response, CCGT-only.

Microstructure $\Rightarrow$ macro. The same finer clock that gives bidders within-hour discrimination also lowers system-level balancing cost.

Thank you.

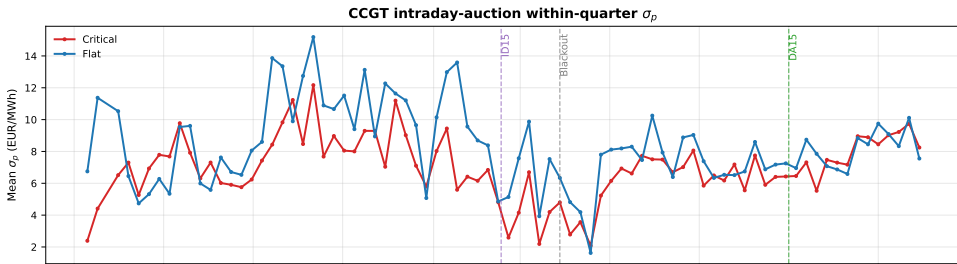
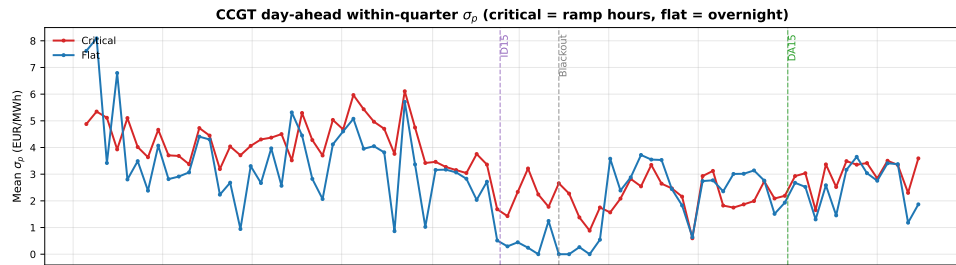
Questions?

Appendix — In-band offered energy migrates to the granular market (Spec B)

Tech	Hour class	ID15 (Net, MWh/class-hr)		DA15 (Net, MWh/class-hr)	
		DA	IDA	DA	IDA
CCGT	morning ramp	77	+2090***	+3462**	−1627***
	midday	−1497***	+1426***	988**	773*
	evening ramp	191	+1859***	+2458***	−152
	flat	−863**	+1405***	+2966**	−1157***
Pump-storage	midday	151	+979**	+433**	+868**
	evening ramp	−1	+775*	251	−408

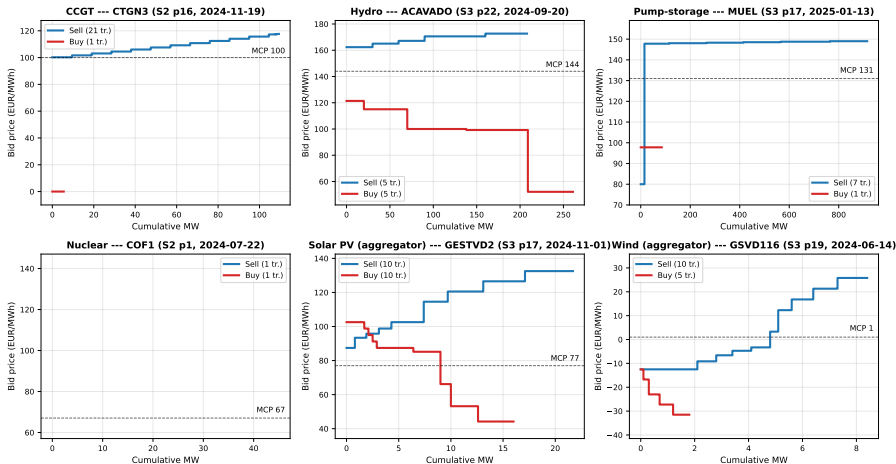
- Under ID15, CCGT and pump-storage push energy *into* IDA (the granular leg); under DA15, they push it back into DA.
- Hydro is the exception (reservoir-driven, 2024 placebo poor); not leaned on.
- Bid-side primitive of the cleared scale-up in critical hours under DA15.

Appendix — Bid-shape parallel trends (visual check)



Appendix — Representative buy vs sell IDA bid curves

Representative IDA sell vs buy bid curves by technology (pre-ID15)



• Sell curves (blue supply) are typically higher-laddered, including parked tier above MCP

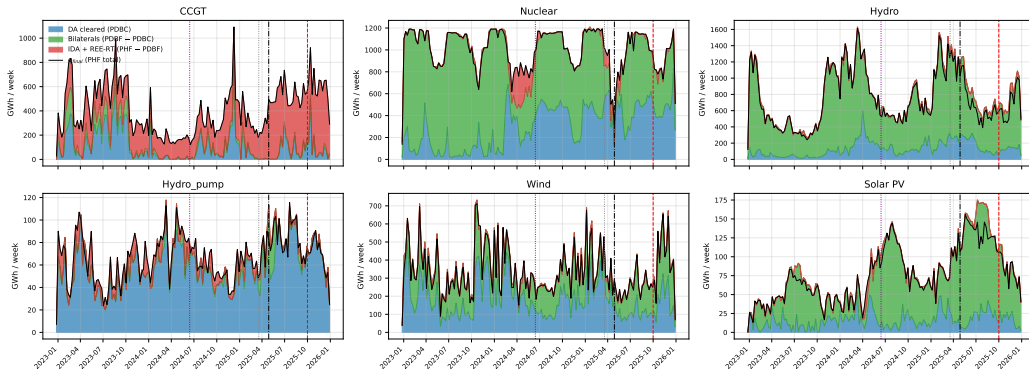
Appendix — Production-unit arbitrageur summary (full sample)

Tech-group	n unit-sessions	% dual	Mean sell MW	Mean buy MW	Buy/Sell
Solar PV	2 093 392	26.9	54	55	101%
Wind	2 313 441	16.2	133	167	126%
Cogen	334 852	3.8	85	92	108%
Biomass	112 548	6.4	27	28	104%
Hydro	339 500	5.7	8 654	3 711	43%
Hydro pump	53 763	4.6	3 462	1 737	50%
CCGT	266 085	4.0	4 625	2 001	43%
Nuclear	13 054	4.1	1 225	360	29%

- VRE / small: buy $\approx$ sell $\Rightarrow$ forecast-driven rebalancing.
- Thermal / storage: buy/sell $\in [23\%, 50\%]$ $\Rightarrow$ sell-dominated rebalancing.
- Arbitrageur active across every tech.

Appendix — Reforzada cost is CCGT-only (post-blackout)

Three-component decomposition (WEEKLY) per technology: DA + Bilaterals + IDA+REE-RT = q_{final} .



- CCGT day-ahead-cleared layer (blue) collapses post-blackout; IDA + REE-RT (red) balloons.
- Other technologies flat across the same window.
- $\Rightarrow$ CCGT-specific intervention, not a uniform reform effect.

Appendix — Robustness summary

- **Pre-trend placebo.** ID15 IDA σ_p clean. DA15 DA: CCGT clean; pump-storage not separable from trend.
- **Cross-border + reforzada.** Hourly cross-border control leaves per-tech DiDs essentially unchanged. Reforzada acts on Fase I level, not within-hour SD.
- **Per-IDA-session BSTS.** ID15 price-drop point estimate near-identical across S1, S2, S3 under short-pre spec (level effect itself not retained — see Result 4).
- **Per-firm.** DA15 CCGT σ_p widening driven by Naturgy and Iberdrola.
- **Offer-type.** Restricting to simple offers, CCGT σ_p DiD rises to 3.53***.
- **2026 placebo (ID15).** Independent 2026 same-calendar placebo confirms wedge SD jump.